

Software Engineering

Organisational Structures

What is a team?

Key Characteristics of a Team

Common Goal: Teams are formed with a shared purpose, which can range from completing a project to achieving a business objective.

Collaboration: Team members work together, leveraging each other's strengths and skills to accomplish tasks more effectively.

Communication: Open and effective communication is essential for teams to coordinate efforts, share information, and make decisions.

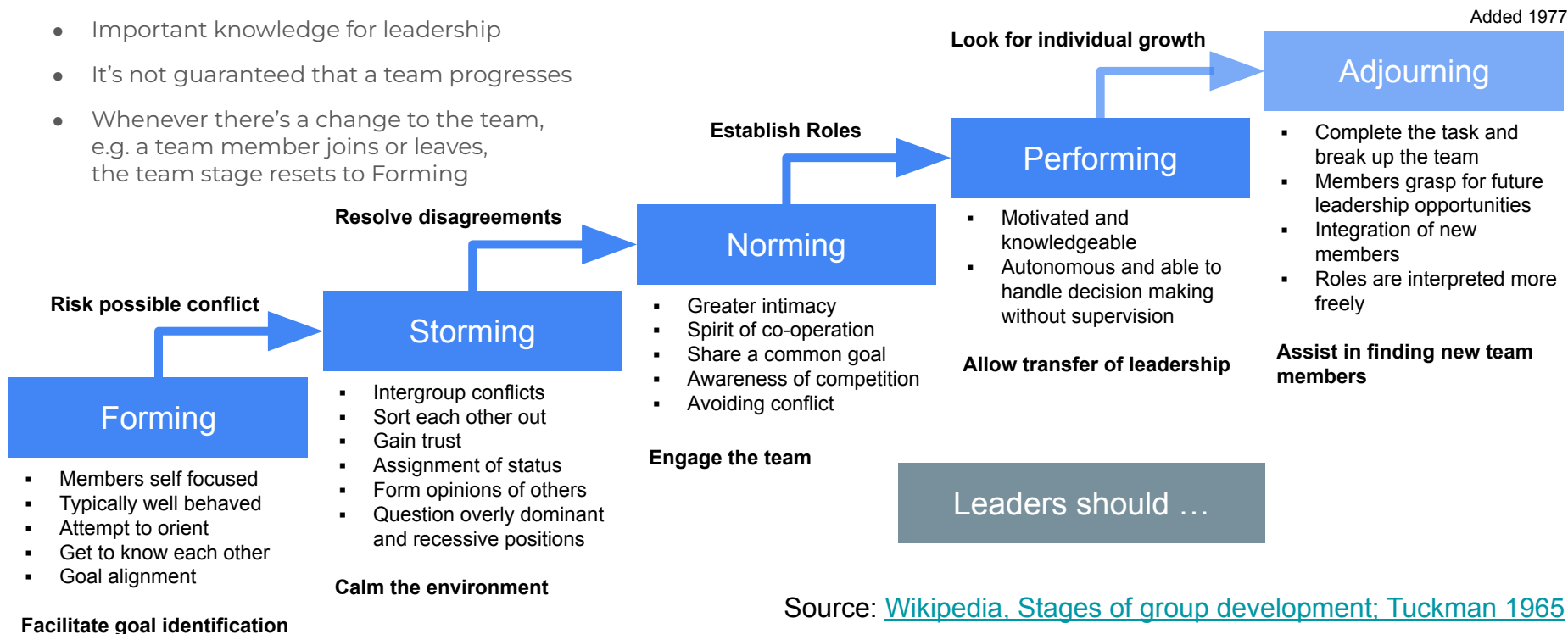
Complementary Skills: Members often have different but complementary skills, allowing the team to tackle complex tasks that require diverse expertise.

Interdependence: Team members depend on each other to complete tasks and achieve the overall goal. Success is typically a collective effort.

Accountability: While the team as a whole is responsible for achieving its objectives, individual members are also accountable for their contributions.

Stages of team development

- Important knowledge for leadership
- It's not guaranteed that a team progresses
- Whenever there's a change to the team, e.g. a team member joins or leaves, the team stage resets to Forming

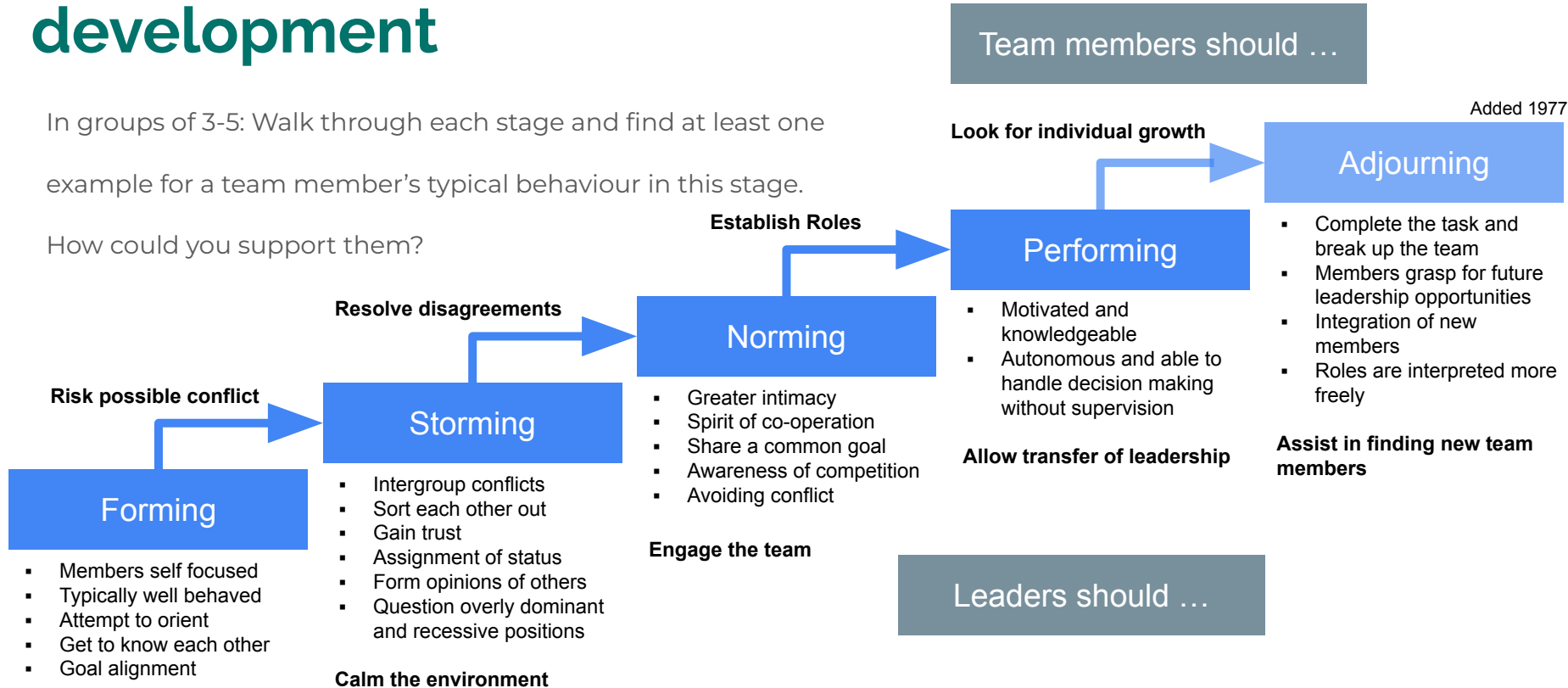


Source: [Wikipedia, Stages of group development; Tuckman 1965](#)

Exercise - 15 minutes: Stages of team development

In groups of 3-5: Walk through each stage and find at least one example for a team member's typical behaviour in this stage.

How could you support them?



Source: [Wikipedia, Stages of group development; Tuckman 1965](#)

Exercise: Team Scenario

Alex, the Lead Developer, carefully chooses words and tries to sound friendly but authoritative. Priya, the UX Designer, is polite and enthusiastic. She checks on other's reactions before fully opening up about her ideas. Marco, the Backend Developer, stays mostly quiet and is taking a lot of notes. Dana, the QA Engineer, asks the usual, procedural questions about workflows and apologises whenever she is understood as being critical. Meanwhile, Ethan, the Scrum Master, gently guides the conversation and emphasises whenever there are common perspectives. Everyone is being extra polite, avoiding potential disagreements.

Matching the behaviour of the team according to Tuckman's stages of team development, what stage is the team in?

What is the next stage and how can you support the team to progress in their team development?

Project OR Product?

Project OR Product?

What is the difference between
a project and a product?

Project or Product?

Project management is the process of supervising the work of a team to achieve all project goals within the given constraints. The primary constraints are scope, time, and budget.

A **project** is a temporary and unique endeavor designed to produce a **product**, service, or result with a defined beginning and end (usually time-constrained, and often constrained by funding or staffing)

The **temporary nature of projects stands in contrast with business as usual** (or operations), which are repetitive, permanent, or semi-permanent functional activities to produce products or services.

Source: [Wikipedia](#)

Product management is the business process of planning, developing, launching, and managing a **product** or service. It includes the **entire lifecycle of a product**, from ideation to development to go to market. Software product management adapts the fundamentals of product management for digital products.

Source: [Wikipedia](#)

In marketing, a **product** is an object, or system, or service made available for consumer use as of the consumer demand. In retailing, products are often referred to as merchandise, and in manufacturing, products are bought as raw materials and then sold as finished goods. A service is also regarded as a type of product.

Source: [Wikipedia](#)

Project OR Product? - Exercise: 5 minutes

Discuss in groups of 2-3:

Is the company you work for
managing digital **projects** or
creating digital **products**?

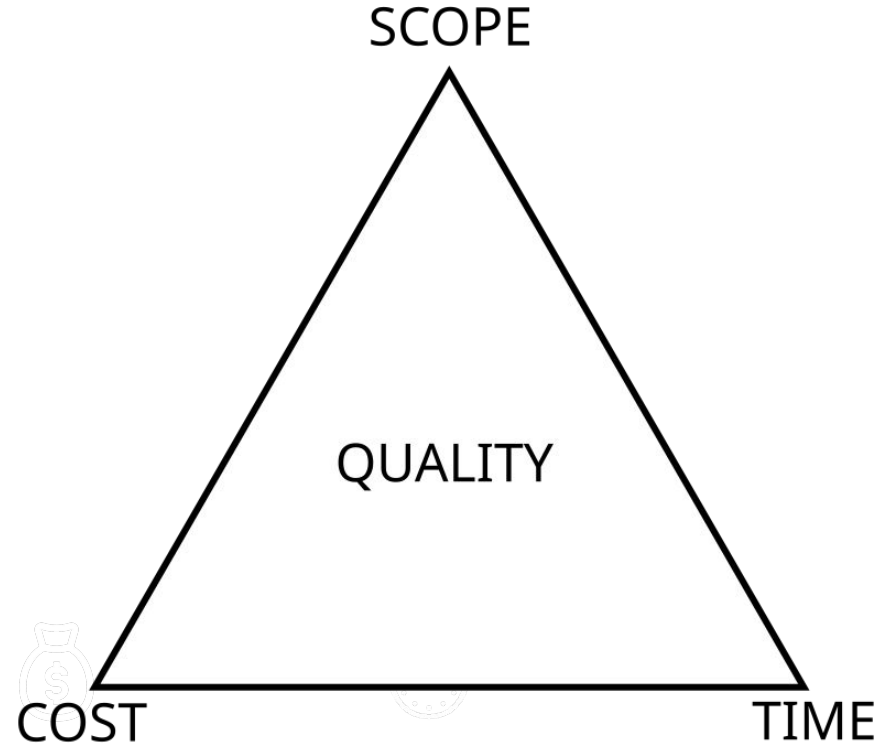
Project Management Triangle

Created in the 1950ies

- The quality of work is constrained by the project's budget, deadlines and scope (features).
- The project manager can trade between constraints.
- Changes in one constraint necessitate changes in others to compensate or quality will suffer.

Source: [Wikipedia](#)

Still helpful for managing products:
Think of product management as an endless loop of managing projects

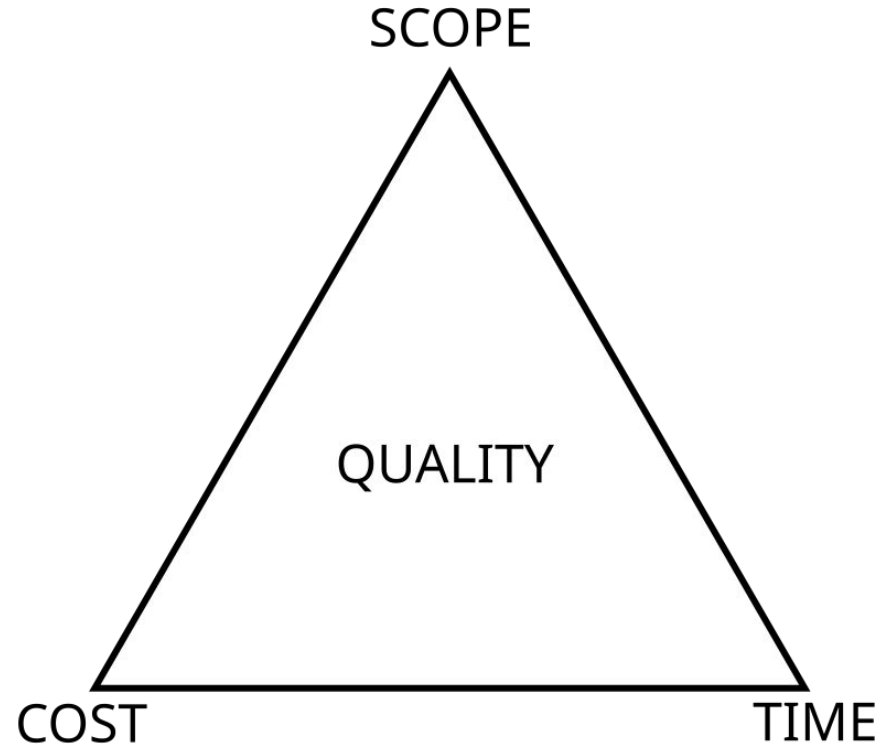


Project Management Triangle - Exercise

We need to build a feature for annual tax declaration that our users need by end of August so that they still have two months to file their taxes.

The estimation exceeds December.

What can we do?

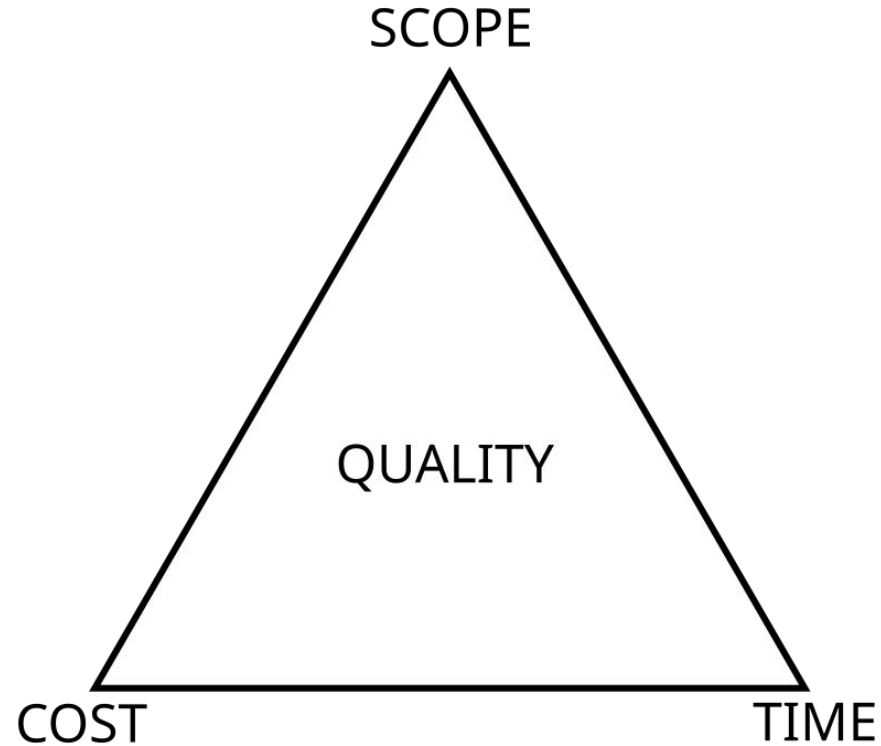


Project Management Triangle - Exercise

We want to add a video editor to our platform. We didn't even manage to implement the most necessary features yet which would bind two of three of our teams. Even then, there is a long list of features that we don't know if the users really need them.

Nevertheless, marketing has promised the release of the video editor in the next two months.

What can we do?

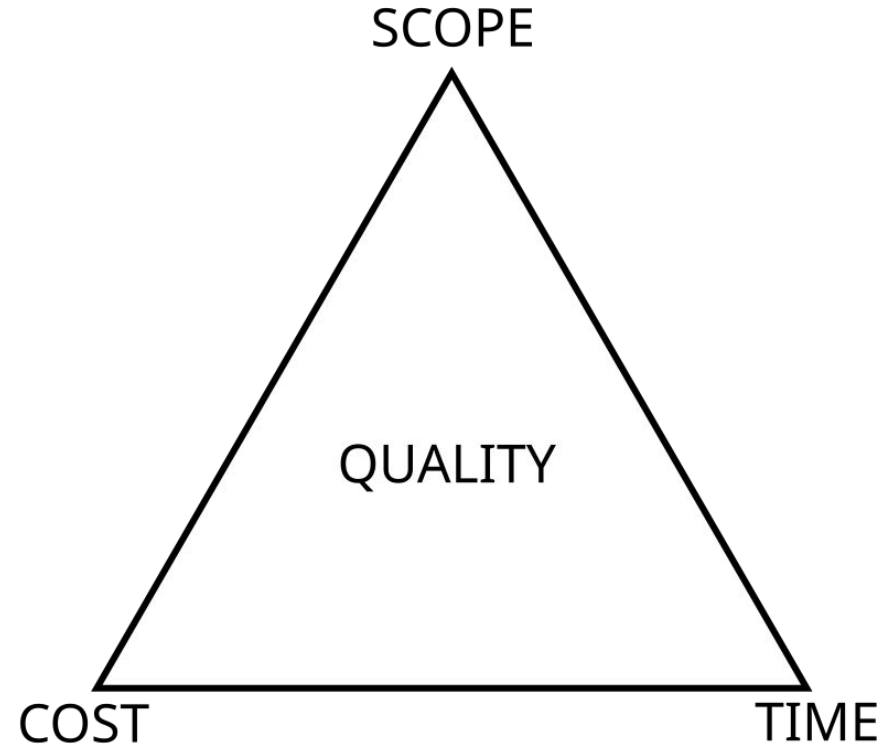


Project Management Triangle - Exercise

We have set the deadline by the end of the week to have an ambitious stretch goal that motivates us.

The understaffed team managed to create all of the features for our product but many are buggy and we don't know the state of our software quality.

What can we do?

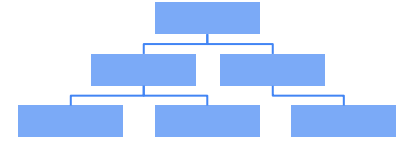


Forms of Organisation

Forms of Organisation

1. Structural organisation – Aufbauorganisation

- Localizes roles and responsibilities
- Establishes a *chain of command* (hierarchy)
- Defines job positions per department (head count)



2. Procedural organisation – Ablauforganisation

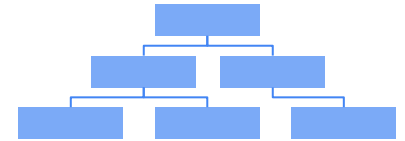
- Establishes overarching processes (e.g. recurring meetings)
- Enables sub-process alignments (e.g. core working times)



Forms of Organisations – Structural Organisation

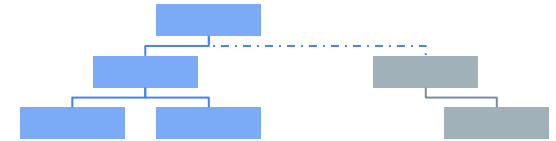
1. Functional organisation:

- Members are grouped by department
- Departments serve a specific function as a unit (line)



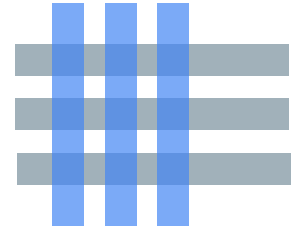
2. Project organisation

- Members are assigned to specific projects
- Projects deliver a solution (in quality, time and budget)
- Members leave their units to work on projects



3. Matrix organisation

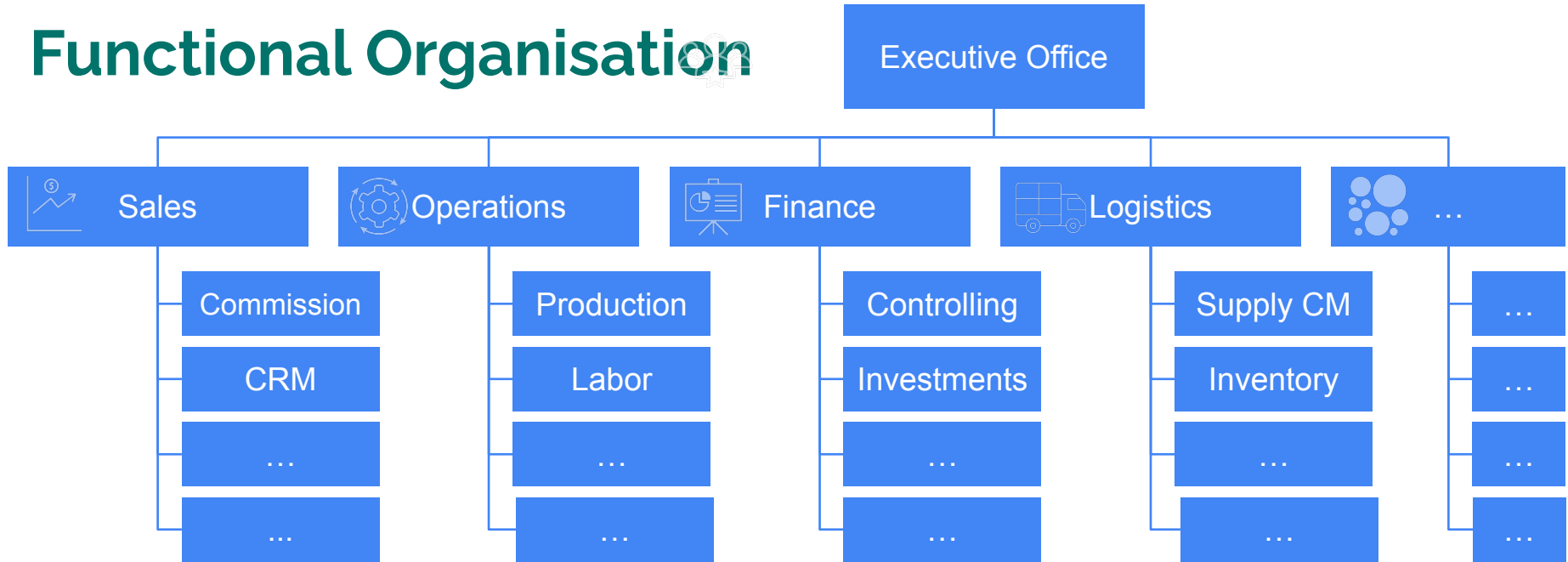
- Members are part of different functional units
- They are assigned to (multiple) teams while staying within their units



Organisational Position

Team Position

Functional Organisation



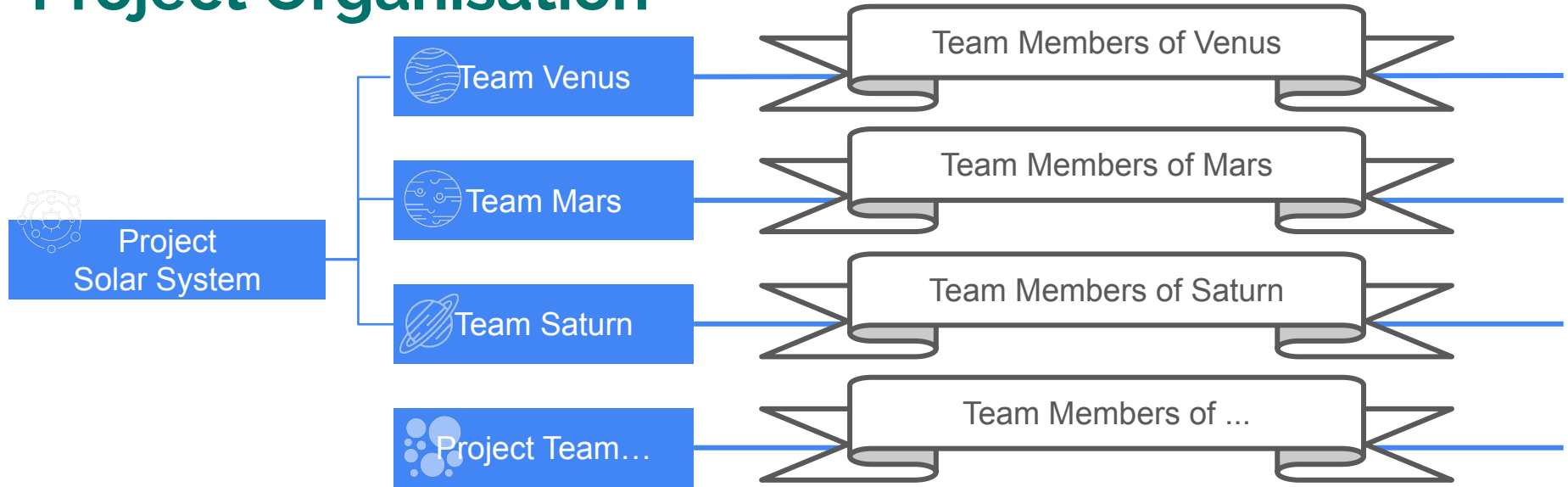
Advantages

- Clear business unit separation
- Shared domain expertise enables knowledge sharing within the departments

Disadvantages

- Projects have to move through lines to complete
- Conway's law will prevent communication across departments

Project Organisation



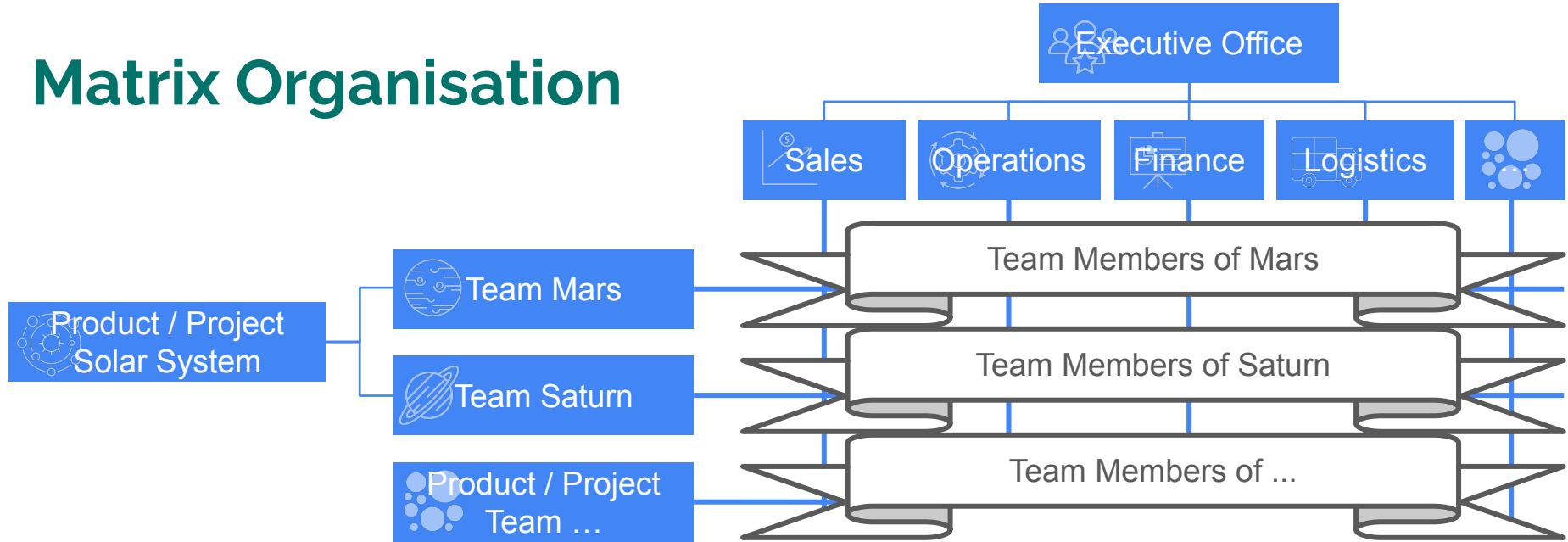
Advantages

- Less idle time as there is no handover across the structure
- Quick integration of new and external members

Disadvantages

- Projects need to ramp up
- As this is a new structure roles, responsibilities and processes have to be defined

Matrix Organisation



▪ Advantages

- Same as project organisation
- Additional long term alignment on line organisation

▪ Disadvantages

- Same as project organisation
- Line lead and project lead have to align. Can lead to ambiguity in priorities

An example of a matrix organisation: The spotify model

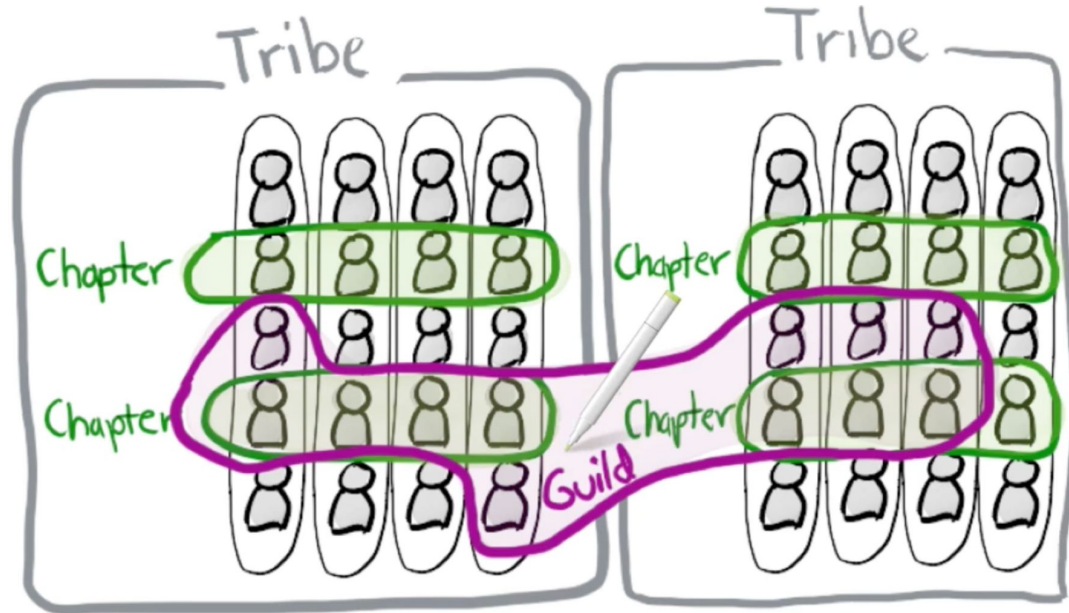
Squads: Autonomous End2End responsibility of an independent user need (search, monitoring)

Chapters: Competency area (QA, Coaching, Development)

Tribes: Lightweight matrix over Chapters and Squads (Region, Application, Process)

Guilds: Loose organisation to share and develop capabilities

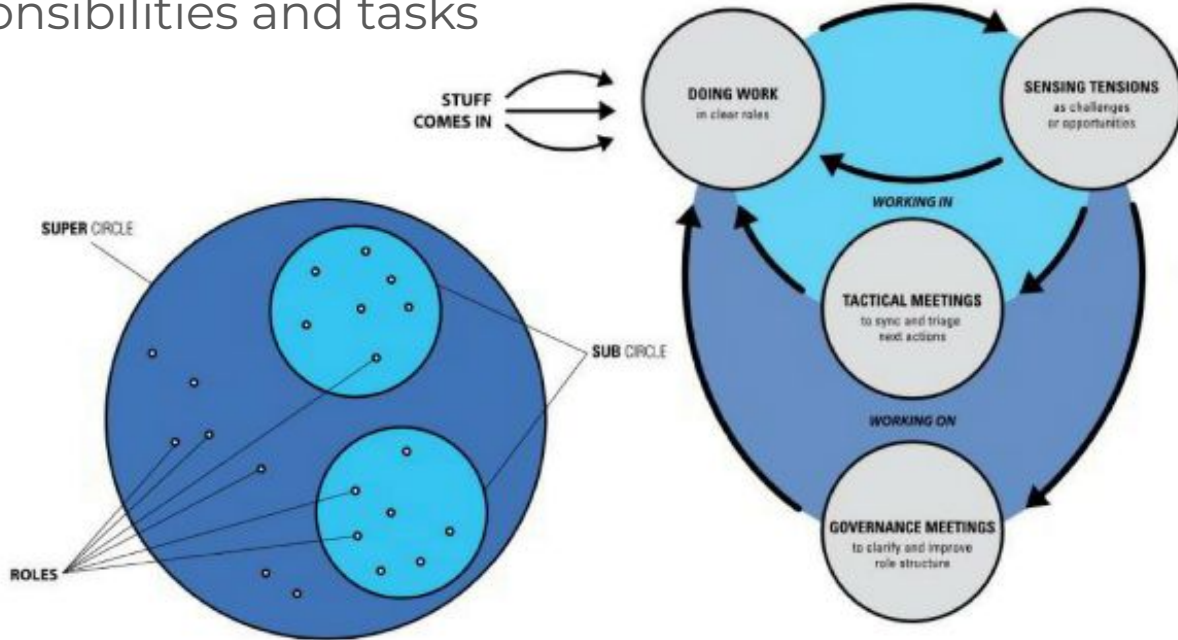
Source: [Spotify Engineering Culture](#)



Other organisation forms: Holacracy

- Super-Circle (Corporation)
- Divided in to sub-circles as autonomous functional units
- Roles have specific responsibilities and tasks
- Governance roles connect super- and sub-circles

Further reading:
holacracy.org



Further Reading

Team Topologies by
Matthew Skelton and Manuel Pais

